

[55]

SARDAR PATEL UNIVERSITY**US02CBCA01 : Advanced C Programming & Intro. To Data Structure****F.Y.B.C.A.(2nd Sem.) Examination****Date : 11/1/2020****Time : 2.00 – 5.00 pm****Marks: 70****Q-1**

Select the appropriate option for the following questions.

[10]

1. Given the following declarations:

int x; double d; int *p; double *q;

Which of the following expression is allowed?

(A) p = &x; (B) q = &x; (C) p = &d; (D) p = x;

2. Which of the following is not a C memory allocation function?

(A) malloc() (B) calloc() (C) realloc() (D) alloc()

3. Which of the following allows a portion of memory to be shared by different types of data?

(A) Array (B) Union (C) Structure (D) File

4. f = fopen(filename, "r");

Referring to the code above, what is the proper definition for the variable f?

(A) FILE f; (B) struct FILE f; (C) int f; (D) FILE *f;

5. A data structure in which insertion and deletion of an elements occurs at only one end is known as _____.

(A) Queue (B) Tree (C) Stack (D) Graph

6. The term "push" and "pop" is related to the?

(A) Array (B) Stack (C) Queue (D) Linked List

7. A linked list is _____ type of data structure?

(A) Linear (B) Non-Linear (C) Both (A) (D) None
and (B)

8. A data structure in which insertion and deletion of an elements occurs at both the end is known as _____.

(A) Stack (B) Queue (C) Priority Queue (D) Deque

9. Which of the following is an operation of a Stack data structure?

(A) Peep (B) Top (C) Bottom (D) None

10. Which of the following is not a derived data type?

(A) Arrays (B) Pointers (C) Structure (D) Float

Q-2

Do as directed. (Attempt ANY TEN)

[20]

1. Define: indirection operator, pointer variable.
2. List out benefits of pointers.
3. Differentiate between '*' and '&' operators in pointers.
4. Write down the syntax to open the text file both in read and write mode.
5. Explain the fclose() function with example.
6. Differentiate '.' and '->' operators.
7. Draw the Hierarchical Structure of Data Structure.
8. Define: Data Structure. List out different applications of data Structure.
9. Differentiate: primitive and non primitive data structure.
10. What is a Singly Linked list?
11. What is a doubly Linked list?
12. State various types of queue.

Q-3

Write a note on Dynamic memory allocation.

[10]

OR

Q-3

Define pointer variable. How can we declare and initialize pointer variable? How can we access value of variable through pointer type variable?

[10]

Q-4

(A) Write note on: pointer to structure

[6]

(B) Write note on: structure within structure

[4]

OR

Q-4

(A) Explain array of structures using suitable examples.

[5]

(B) Explain the all the modes of file management with example.

[5]

Q-5

Explain the linear and non linear data structure briefly.

[10]

OR

Q-5

Explain a STACK with an example. Write along with for various operations performed over a stack.

[10]

Q-6

(A) Write an algorithm to insert an element at the ending of a Singly linked list.

[5]

(B) Write an algorithm to delete an element from a Singly linked list.

[5]

OR

Q-6

Write an algorithm to insert and delete of the element from a queue.

[10]

— X —